

LAMPANG AGROMET, ThailandWMO: **483340**Lat: **18.3250N**Lon: **99.3014E**Elev: **1047**StdP: **14.15**Time Zone: **7.00 (E07)**Period: **08-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	55.6	58.6	48.7	52.9	71.4	51.0	57.6	70.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	22.8	101.5	75.0	98.8	74.9	96.5	74.5	80.8	89.2	80.1	88.3	79.8	87.7	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.9	156.2	84.2	78.3	152.7	83.6	77.9	150.6	83.2	45.5	89.9	44.9	88.7	44.3	87.9	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	79.8	72.3	76.6	82.0	85.9	84.2	82.8	81.9	81.1	81.2	79.6	77.5	72.6
	DBStd	5.15	3.76	3.30	3.92	3.99	3.35	2.25	2.58	1.67	1.82	2.08	2.53	4.25
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	12	5	0	1	0	0	0	0	0	0	0	0	6
	CDD50	10883	690	744	991	1078	1060	984	990	965	936	919	826	700
	CDD65	5419	230	324	526	628	595	534	525	500	486	454	376	241
	CDH74	55918	1849	3471	6448	8254	6740	5739	5336	4658	4632	3932	3107	1752
	CDH80	23793	537	1628	3572	4807	3205	2218	1932	1460	1574	1336	1060	464
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(15) Precipitation	PrecAvg	43.9	0.4	0.4	1.1	2.6	6.8	4.8	5.7	8.0	8.3	4.2	1.3	0.4
	PrecMax	66.6	2.8	1.9	4.6	6.7	12.0	8.0	9.5	14.1	15.7	7.2	6.1	3.8
	PrecMin	30.6	0.0	0.0	0.0	0.4	1.0	1.5	2.9	3.3	4.1	1.2	0.0	0.0
	PrecStd	7.7	0.7	0.5	1.2	1.5	3.1	1.7	1.9	2.6	2.4	1.6	1.3	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.3	95.8	101.0	105.8	101.8	96.3	97.1	92.1	92.6	91.2	90.4	88.7
		MCWB	70.1	69.1	71.8	74.5	77.8	79.2	78.0	78.6	79.9	78.4	76.9	74.3
	2%	DB	87.0	93.6	98.8	103.4	98.2	93.5	93.3	90.1	90.8	89.5	88.9	86.5
		MCWB	70.2	69.7	71.8	74.9	77.8	78.9	78.0	78.6	79.5	78.2	76.0	73.5
	5%	DB	85.1	91.6	96.9	100.8	95.3	91.6	90.9	88.7	89.2	88.3	87.5	84.7
		MCWB	70.1	70.4	71.2	75.0	78.0	78.9	78.3	78.5	79.2	77.8	75.5	72.7
	10%	DB	83.3	89.4	94.8	98.0	92.8	89.6	88.6	87.0	87.7	86.9	86.0	82.8
		MCWB	69.6	70.1	71.5	75.5	78.2	78.7	78.3	78.4	78.7	77.5	75.4	71.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.9	75.6	77.1	79.6	81.5	81.1	81.0	80.9	81.5	80.1	78.9	77.4
		MCDB	83.1	87.7	92.1	95.0	91.9	90.2	88.2	87.6	89.1	87.4	86.9	84.2
	2%	WB	74.0	74.4	75.8	78.3	80.3	80.4	80.1	80.0	80.6	79.5	77.6	75.9
		MCDB	81.4	86.7	90.3	94.1	90.7	89.3	87.4	86.5	88.3	86.6	85.2	83.0
	5%	WB	72.6	73.3	74.8	77.4	79.6	79.7	79.4	79.4	79.9	78.8	77.0	74.3
		MCDB	80.4	85.6	89.1	92.8	89.8	88.0	86.8	85.8	87.1	85.6	84.5	81.5
	10%	WB	71.2	72.0	73.8	76.6	78.9	79.1	78.8	78.9	79.2	78.1	76.4	73.0
		MCDB	79.4	83.7	88.4	91.5	89.1	86.7	86.2	85.1	85.8	84.6	83.2	80.2
(35) Mean Daily Temperature Range	5% DB	MDBR	22.4	27.3	25.8	22.8	16.6	13.3	12.7	12.2	13.2	14.1	17.5	20.2
		MCDBR	25.3	30.0	28.9	26.7	20.7	16.7	16.9	14.9	15.7	16.5	18.7	21.5
		MCWBR	10.3	10.3	8.9	6.7	5.4	4.8	4.9	5.2	6.1	6.6	7.4	9.6
	5% WB	MCDBR	19.3	24.8	24.2	22.3	17.5	14.4	13.8	13.0	14.7	14.6	16.5	18.6
		MCWBR	9.0	10.0	8.1	6.8	6.0	5.1	5.5	5.6	6.5	6.3	6.9	9.0
(40) Clear-Sky Solar Irradiance	taub		0.436	0.514	0.651	0.642	0.493	0.434	0.423	0.437	0.470	0.515	0.439	0.434
	taud		2.201	1.996	1.682	1.763	2.202	2.398	2.427	2.371	2.259	2.088	2.268	2.232
	Ebn at Noon		266	251	222	226	259	272	275	273	265	248	264	263
	Edh at Noon		43	56	79	73	47	38	37	39	44	50	40	41
(44) All-Sky Solar Radiation	RadAvg		1453	1683	1781	1914	1726	1518	1417	1399	1460	1425	1462	1374
	RadStd		73	63	114	108	148	115	103	89	65	81	98	75

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)	+0.49	N/A	N/A	N/A	N/A	N/A	+0.27	N/A	N/A	+225	+227

Nomenclature: See separate page